Project Document: Handwritten Digit Recognition (MNIST on Kaggle)

📝 Problem Statement

In the modern digital era, recognizing handwritten digits has numerous applications, such as digitizing bank cheques, reading postal codes, or developing assistive technologies.
Our problem is:
"Can we build a machine learning model that can correctly identify handwritten digits (0–9) from images?"

📊 Dataset (Kaggle: Digit Recognizer Competition)

- **Source:** Kaggle Digit Recognizer
- **Training Data:** `train.csv`
    - Contains **42,000 images** of digits (28×28 pixels).
    - Each image is labeled with the correct digit (0–9).
- **Test Data:** `test.csv`
    - Contains **28,000 images** without labels (we have to predict them).
- **Image Format:** Flattened 28×28 grayscale images (total 784 pixel values per row).

🛠️ Tools and Libraries Used

We will use the following technologies:

- **Python** → Main programming language.
- **Pandas & NumPy** → Data loading, handling, and manipulation.
- **Matplotlib & Seaborn** → Visualization of digits and results.
- **Scikit-learn** → Train/validation split, confusion matrix, metrics.
- **TensorFlow/Keras** → Build and train a **Convolutional Neural Network (CNN)**.

🤖 Algorithms & Approach

1. **Data Preprocessing**
    - Normalize pixel values (0–255 → 0–1).
    - Reshape images to 28×28 for CNN.
    - One-hot encodes the labels.
2. **Model Selection** → Convolutional Neural Network (CNN)
    - CNNs are very effective for image recognition tasks.
    - Layers used:
        - **Conv2D** → Extract image features.
        - **MaxPooling** → Reduce dimensionality.
        - **Dropout** → Prevent overfitting.
        - **Dense (Softmax)** → Output probabilities for digits (0–9).

3. **Model Training**
    - Train on the training set.
    - Validate with a portion of the data.
    - Use **accuracy** as the metric.
4. **Evaluation**
    - Plot accuracy and loss curves.
    - Confusion matrix to check which digits get misclassified.
5. **Prediction & Submission**
    - Predict on `test.csv`.
    - Save results in `submission.csv` for Kaggle.

✅ Project Conclusion

- We successfully built a **deep learning model (CNN)** that can recognize handwritten digits with **high accuracy (~99%)**.
- This project teaches:
    - How to handle image datasets in tabular format.
    - Basics of CNN architecture for computer vision.
    - End-to-end ML workflow: **problem → data → preprocessing → model → evaluation → deployment**.
- Practical impact: Forms the foundation for advanced computer vision tasks such as face recognition, OCR (Optical Character Recognition), and autonomous systems.